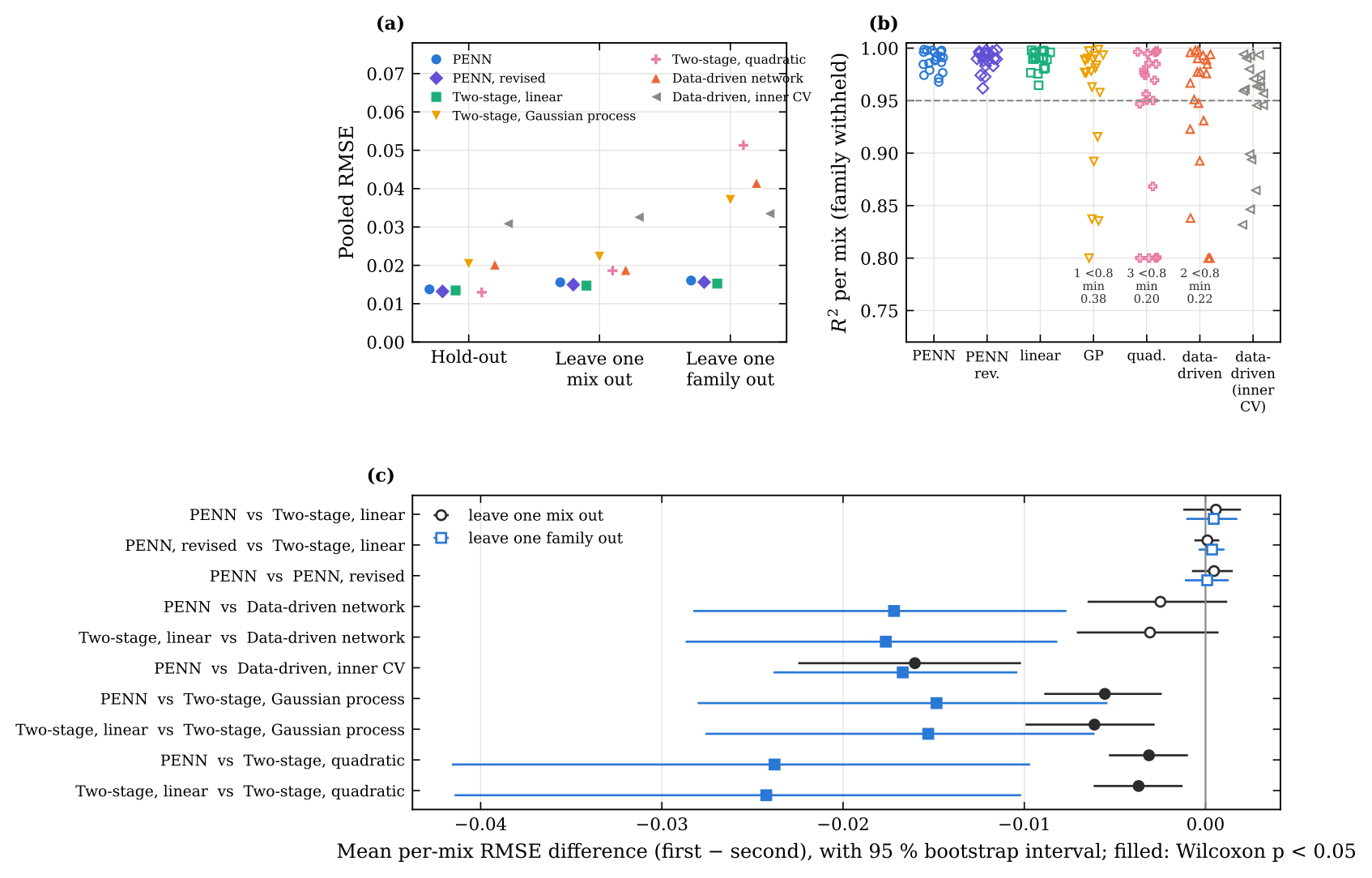
(a)
PENN
PENN, revised
Two-stage, linear
Two-stage, Gaussian process
Two-stage, quadratic
Data-driven network
Data-driven, inner CV
Pooled RMSE
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
Hold-out
Leave one mix out
Leave one family out
(b)
R^2 per mix (family withheld)
0.75
0.80
0.85
0.90
0.95
1.00
1 <0.8 min 0.38
3 <0.8 min 0.20
2 <0.8 min 0.22
PENN
PENN rev.
linear
GP
quad.
data-driven
data-driven (inner CV)
(c)
leave one mix out
leave one family out
PENN vs Two-stage, linear
PENN, revised vs Two-stage, linear
PENN vs PENN, revised
PENN vs Data-driven network
Two-stage, linear vs Data-driven network
PENN vs Data-driven, inner CV
PENN vs Two-stage, Gaussian process
Two-stage, linear vs Two-stage, Gaussian process
PENN vs Two-stage, quadratic
Two-stage, linear vs Two-stage, quadratic
−0.04
−0.03
−0.02
−0.01
0.00
Mean per-mix RMSE difference (first − second), with 95 % bootstrap interval; filled: Wilcoxon p < 0.05